Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 110405

Subject: Microprocessors

Time: 03 Hours
Full Marks: 70

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The default stack pointer value after reset in the 8051 is
(i) 00H
(ii) 07H
(iii) 08H
(iv) FFH

(b) What will be the final value of Accumulator after execution?

```
MOV A,#55H
ANL A,#0FH
ORL A,#A0H
```

(i) A5H
(ii) F5H
(iii) 55H
(iv) A0H

(c) How many bytes are required for MOV A,#25H?
(i) 1
(ii) 2
(iii) 4
(iv) 3

(d) An 8-bit microprocessor with a 16-bit address bus can access a maximum memory of
(i) 8 KB
(ii) 16 KB
(iii) 64 KB
(iv) 128 KB

(e) Memory-mapped I/O uses
(i) Separate address space
(ii) Same address space as memory
(iii) Stack memory
(iv) Program memory only

(f) Which register points to the top of the stack?
(i) PC
(ii) DPTR
(iii) SP
(iv) ACC

(g) ZigBee is based on
(i) IEEE 802.3
(ii) IEEE 802.11
(iii) IEEE 802.15.4
(iv) IEEE 802.16

(h) What is the output of the following code?

```
MOV A,#0AAH
XRL A,#0FFH
```

(i) 55H
(ii) AAH
(iii) FFH
(iv) 00H

(i) SPI is a
(i) Single-wire protocol
(ii) Parallel protocol
(iii) Full-duplex serial protocol
(iv) Wireless protocol

(j) Stepper motors move in
(i) Continuous rotation
(ii) Random motion
(iii) Discrete steps
(iv) Linear motion

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Q.2 (a) Explain memory-mapped I/O versus isolated I/O techniques. [7]

(b) Show the status of the CY, AC, and P flags after the addition of 88H and 93H in the following instructions. [7]

MOV A, #88H

ADD A, #93H

Q.3 (a) Explain the TMOD and TCON registers in the 8051 microcontrollers. [7]

(b) Discuss the machine cycle, instruction cycle, and state transitions in the 8051. [7]

Q.4 (a) Explain the role of microcontrollers in embedded systems and justify why microcontrollers are preferred over microprocessors in embedded applications. [7]

(b) Explain the importance of $\overline{PSEN}$ and $\overline{EA}$ pins of the 8051 microcontrollers. [7]

Q.5 (a) Explain the architecture of 8051. [7]

(b) Discuss synchronization issues during peripheral interfacing. [7]

Q.6 (a) Find the sum of the values 79H, F5H, and E2H. Put the sum in registers R0 (low byte) and R5 (high byte). [7]

(b) Analyze ZigBee protocol features for low-power embedded systems. [7]

Q.7 (a) Explain DC motor interfacing using drivers and PWM control. [7]

(b) Explain the RS-232 protocol architecture and evaluate its limitations. [7]

Q.8 (a) Explain the interfacing of external RAM and ROM with microcontrollers. [7]

(b) Write an 8051 assembly language program to transfer a block of 10 bytes from internal RAM starting at address 30H to external RAM starting at 2000H using indirect addressing. Mention the addressing modes used. [7]

Q.9 Write short note on *any two* of the following: [7 × 2 = 14]

(a) Flag bits and PSW register

(b) Stack and Stack Pointer

(c) Compare microprocessor and microcontroller

(d) Addressing modes

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